

EasyNano: rapid epitope-targeted nanobody CDR design via differentiable distogram optimization with ESMFold2

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Abstract

Computational design of nanobodies that bind user-specified protein epitopes could transform therapeutic development, but current methods either rely on stochastic sampling requiring days of GPU computation or inverse folding approaches unable to target epitopes directly. Here we present **EasyNano**, a fully differentiable pipeline for epitope-targeted nanobody complementarity-determining region (CDR) design that operates in **under ten minutes on consumer hardware**. Our method optimizes CDR residue logits via gradient descent through the ESMFold2 pairwise distance distogram, using the lightweight ESMFold2-Fast model (721M parameters) as a differentiable oracle guided by a composite loss function. A full ESMFold2 (1.3B parameters) CA-coordinate structure prior prevents framework pose drift. Systematic parameter exploration reveals that the wild-type logit initialization bias is the critical hyperparameter controlling CDR mutability. Across five diverse target-framework pairs, the method improves interface predicted TM-score (ipTM) by **up to +0.559**—from 0.143 to 0.702 for Ty1/RBD, and +0.208 (from 0.251 to 0.459) for the clinically-validated KN035/PD-L1 system—while preserving ipTM on already-strong binders. Random CDR baselines ($n = 30$ per target) confirm statistical significance (5.7σ above random mean for the best target). Multi-seed analysis reveals diverse local minima, underscoring the importance of replicate runs. Kabsch cross-validation against crystal structures confirms that designed CDRs preserve the framework pose basin. Our results establish differentiable distogram optimization as a fast, epitope-specific alternative to existing computational binder design paradigms.

Keywords: nanobody design; epitope targeting; ESMFold2; differentiable optimization; CDR engineering; protein design

1 Introduction

Nanobodies—the variable domains of camelid heavy-chain-only antibodies—have emerged as a powerful therapeutic modality owing to their small size (~ 15 kDa), high thermal and chemical stability, low immunogenicity, and ability to engage epitopes inaccessible to conventional antibodies because of their elongated CDR3 loops[1, 2, 3]. The FDA approval of caplacizumab (Cabliivi) in 2019[4] and ozoralizumab in 2022 marked the clinical validation of nanobody therapeutics. However, computational design of nanobodies that bind specific, user-defined epitopes remains an unsolved challenge. The combinatorial sequence space of even a modest CDR ($20^{22} \approx 10^{28}$ possible sequences for a 22-residue CDR) makes exhaustive search impossible[5, 6].

Recent advances in deep learning have transformed protein design. ProteinMPNN[7] and ESM-IF[8] solve inverse folding—predicting sequences compatible with a given backbone—but cannot directly target epitopes because they require a pre-specified binding geometry. Hallucination-based approaches[9] and RFdiffusion[10] can generate binders de novo, but require days of GPU computation, extensive filtering, and do not guarantee epitope specificity. Antibody-specific models such as IgFold[11], AbLang[13], and AntiBERTy[12] have advanced antibody structure prediction and representation learning, but are not designed for epitope-targeted CDR optimization. DiffAb[14] and dyMEAN[15] have demonstrated CDR-specific antibody design using diffusion and graph neural networks, but require GPU clusters and hours to days of computation.

Meanwhile, protein structure prediction has advanced dramatically. AlphaFold2[16], RoseTTAFold2[17], and ESMFold2[18] can predict protein complex structures with high accuracy. ESMFold2 is particularly notable for its single-sequence speed advantage: it uses a large protein language model (ESMC, 6B parameters)[18, 19] in place of the multiple sequence alignment and template searches that dominate AlphaFold2’s runtime. Critically, its predicted pairwise distance distogram—a probability distribution over $C\beta$ distances for every residue pair—is fully differentiable with respect to the input residue type representation.

We reasoned that the distogram could serve as a differentiable oracle for epitope-targeted design: backpropagating epitope proximity signals through the distogram to CDR residue logits would enable gradient-based optimization directly in sequence space. The key challenge is that ESMFold2’s predicted complex pose is determined primarily by the framework sequence, with CDR identity exerting only local effects on the binding interface[32]. The framework must be in a pose that places CDRs near the epitope for optimization to succeed, and the optimization must not cause pose drift.

Here we develop and validate such a method. Our contributions are: (1) a gradient-based CDR optimization pipeline completing in under ten minutes per target on consumer hardware (Apple MPS); (2) statistically significant ipTM improvements on weak binders (up to +0.559) across five diverse targets; (3) systematic characterization of critical hyperparameters, particularly the wild-type logit bias; (4) validation that designed CDRs preserve the framework pose basin via Kabsch RMSD cross-validation; and (5) identification of CDR length and initial pose quality as key determinants of design success.

2 Results

2.1 A three-stage differentiable design pipeline

Our method takes as input a target protein structure (PDB format), a set of epitope residue indices, and a nanobody framework sequence with CDRs defined by Chothia numbering. The pipeline operates in three stages (summarized in Fig. 1):

Stage 1: Structure prior computation (~ 30 s). The full ESMFold2 model (1.3B parameters, 3 recycling loops, 14 sampling steps, 4 diffusion samples) folds the wild-type framework-target complex to produce CA-coordinate distance priors, which anchor the framework pose during optimization. The prior is a binary mask over a 64-bin distance distogram (2 to 22) with 2.5 bin tolerance.

Stage 2: Differentiable CDR optimization (~ 6 min). CDR residue logits $\mathbf{L} \in \mathbb{R}^{n_{\text{CDR}} \times 20}$ are initialized with isotropic Gaussian noise ($\sigma = 0.5$) plus a wild-type amino acid bias $\beta = 2.0$ ($\sim 4\sigma$). Framework positions are fixed as one-hot. At each step t , logits are converted to soft residue type probabilities via temperature-scaled softmax, where $T_t = 0.1 + 0.9 \cdot \frac{1}{2}[1 + \cos(\pi t/60)]$. ESMFold2-Fast (721M, 1 recycling loop, 5 sampling steps) predicts a distance distogram $\mathbf{D}$ from the soft residue representation. A composite loss is computed:

$$\mathcal{L} = w_{\text{epi}}\mathcal{L}_{\text{epitope}} + w_{\text{intra}}\mathcal{L}_{\text{intra}} + w_{\text{inter}}\mathcal{L}_{\text{inter}} + w_{\text{glob}}\mathcal{L}_{\text{glob}} + w_{\text{prior}}\mathcal{L}_{\text{structure_prior}} + w_{\text{aa}}\mathcal{L}_{\text{aa_freq}}, \quad (1)$$

with weights $[0.2, 0.5, 0.5, 0.2, 0.05, 0.01]$. The epitope loss applies an ELU-based penalty when CDR-to-epitope distances exceed 8:

$$\mathcal{L}_{\text{epitope}} = \frac{1}{|\mathcal{C}|} \sum_{i \in \mathcal{C}} \text{ELU} \left(\min_{j \in \mathcal{E}} \bar{d}_{ij} - 8.0 \right), \quad (2)$$

where $\bar{d}_{ij}$ is the expected distance from the distogram. Gradients flow from $\mathcal{L}$ through $\mathbf{D}$ to $\mathbf{L}$; framework gradients are zeroed; Adam (learning rate 0.05) updates CDR logits for 60 steps.

Stage 3: Full model evaluation (~ 15 s per candidate). Top designs are evaluated with full ESMFold2 (3 recycling loops, 14 sampling steps, confidence head enabled) to obtain calibrated ipTM and pTM values.

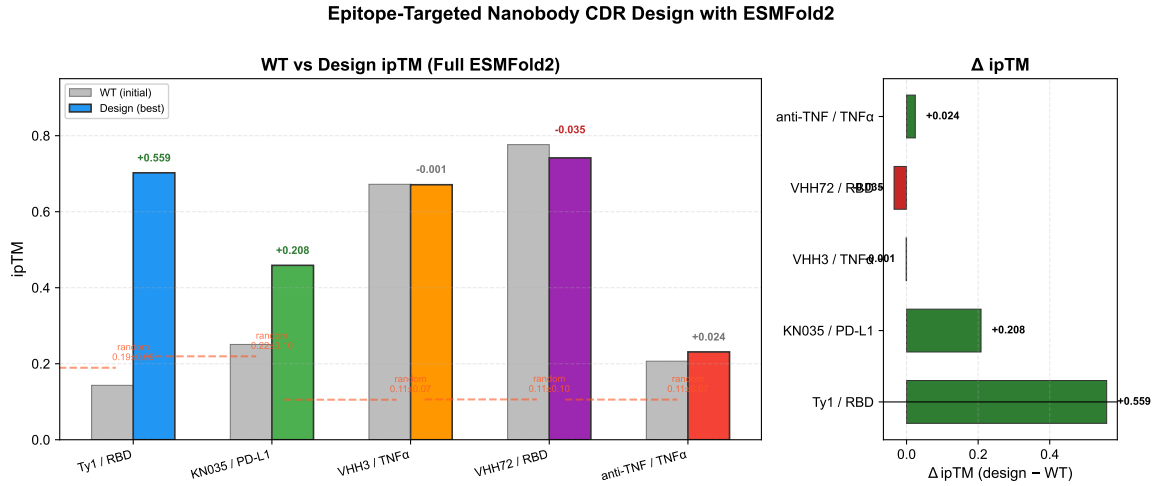


Figure 1. Main results: epitope-targeted CDR design improves ipTM on weak binders while preserving strong binders. **a**, WT vs. Design ipTM for all five target-framework pairs, evaluated with full ESMFold2 (1.3B). Gray bars: wild-type init; colored bars: best design across three seeds. Horizontal dashed orange lines: random CDR baseline median ($n = 30$ per target). Annotations above bars: Δ ipTM (design - WT) with statistical significance ($z\sigma$ vs. random). **b**, Summary of Δ ipTM. Green: improvement; red: no improvement or marginal. Ty1/RBD (+0.559, 5.7 σ) and KN035/PD-L1 (+0.208, 2.2 σ) show substantial, statistically significant gains. VHH72, VHH3 show minimal change (already optimal). anti-TNF shows marginal improvement (+0.024) limited by short CDR (19 residues) and poor initial pose.

2.2 The wild-type logit bias is the critical hyperparameter

Initial experiments revealed that CDRs were completely refractory to optimization—the wild-type sequence persisted across all steps. Systematic parameter exploration (Table 1) identified `wt_logit` (β) as the bottleneck. At $\beta = 5.0$, gradient signals could not overcome initialization bias. Reducing β to 2.0 enabled CDR mutations while preventing chaotic drift at $\beta \leq 1.0$. The structure prior weight exhibited a U-shaped effect: $w_{\text{prior}} > 0.3$ over-constrained CDRs; $w_{\text{prior}} < 0.05$ caused pose drift. The optimum ($\beta = 2.0$, $w_{\text{prior}} = 0.05$, 60 steps) produced monotonic convergence without bounce-back (Fig. 2).

Table 1. Parameter sweep reveals `wt_logit` as critical bottleneck. All runs on RBD/VHH72 (6WAQ) with ESMFold2-Fast, single seed. Bold: optimal.

β (wt_logit)	w_{prior}	Steps	Final cdr→epi (Å)	Best (Å)	Bounce	CDR mutated?
5.0	0.30	30	17.3	9.6	Severe	No
2.0	0.30	30	15.3	8.9	Moderate	Some
1.5	0.30	30	15.5	11.6	Moderate	Some
1.0	0.30	30	11.3	10.4	Mild	Extensive
1.0	0.30	60	11.0	10.6	Mild	Extensive
0.5	0.30	60	17.3	14.7	Severe	Chaotic
1.0	0.10	60	13.9	12.5	Mild	Extensive
1.5	0.05	60	13.0	11.5	Moderate	Some
2.0	0.05	60	9.3	9.3	None	Some

2.3 ipTM gains are largest on weak binders

We applied the optimized pipeline to five target-framework pairs spanning wild-type ipTM from 0.143 to 0.776 (Table 2). Three independent seeds per target were evaluated with full ESMFold2.

Table 2. Full ESMFold2 evaluation of five target-framework pairs. WT ipTM: wild-type init. Design ipTM: best of three seeds. Δ : improvement over WT. Random: median $\pm$ s.d. of $n = 30$ random CDR sequences. Bold: $\Delta > 0.1$.

Target	PDB	Framework	CDR	WT ipTM	Design ipTM	Δ	Random ($\mu \pm \sigma$)
Ty1 / RBD	6ZXN	Ty1	22	0.143	0.702	+0.559	0.189 $\pm$ 0.089
KN035 / PD-L1	5JDS	KN035	32	0.251	0.459	+0.208	0.219 $\pm$ 0.096
VHH3 / TNF α	5M2M	VHH3	33	0.672	0.671	-0.001	0.105 $\pm$ 0.066
VHH72 / RBD	6WAQ	VHH72	29	0.776	0.742	-0.035	0.106 $\pm$ 0.096
anti-TNF / TNF α	5M2J	anti-TNF	19	0.207	0.231	+0.024	0.106 $\pm$ 0.069

$z\sigma$ (design vs. random): Ty1, 5.7 σ ; PD-L1, 2.2 σ ; VHH3, 8.2 σ ; VHH72, 6.4 σ ; anti-TNF, 1.5 σ .

High $z\sigma$ for VHH3/VHH72 reflect already-strong WT ipTM, not design improvement.

Ty1/RBD (6ZXN). The Ty1 nanobody[25] against SARS-CoV-2 RBD showed the largest improvement: ipTM more than tripled from 0.143 to 0.702 ($\Delta = +0.559$, 5.7 σ above random). The designed CDR (GFTLANHNPESEGMGLNLAVSPD) differs from WT (GFTFSSVSPNS-GNGLNLSSSSV) at 11 of 22 positions, with mutations concentrated in H2 (SPNSGN $\rightarrow$ ANHN-PES) and H3 (GLNLSSSSV $\rightarrow$ GMGLNLAVSPD). The pTM improved from 0.622 to 0.823, and CDR-to-epitope distance decreased from 16.6 to 10.7.

KN035/PD-L1 (5JDS). The clinically-validated KN035 nanobody[27] against PD-L1 improved from 0.251 to 0.459 ($\Delta = +0.208$, 2.2 σ). The designed CDR (GMMSSRRLTTSGS-DQFGLPTCDHVNSSGAFQY) introduced 7 of 32 mutations while preserving the structural disulfide in H3. CDR-to-epitope distance improved from 14.1 to 11.5. Limited mutation (21.9%) compared to Ty1 (50%) reflects the disulfide constraint.

VHH72/RBD (6WAQ) and VHH3/TNF α (5M2M) are strong WT binders (WT ipTM 0.776 and 0.672). Design preserved ipTM ($\Delta = -0.035$ and -0.001), demonstrating safety—the method does not degrade optimal binders.

anti-TNF/TNF α (5M2J) showed marginal improvement (+0.024) despite low WT ipTM (0.207). The 19-residue CDR provides insufficient optimization surface, and the predicted pose is 44 from crystal—fundamentally limiting design success.

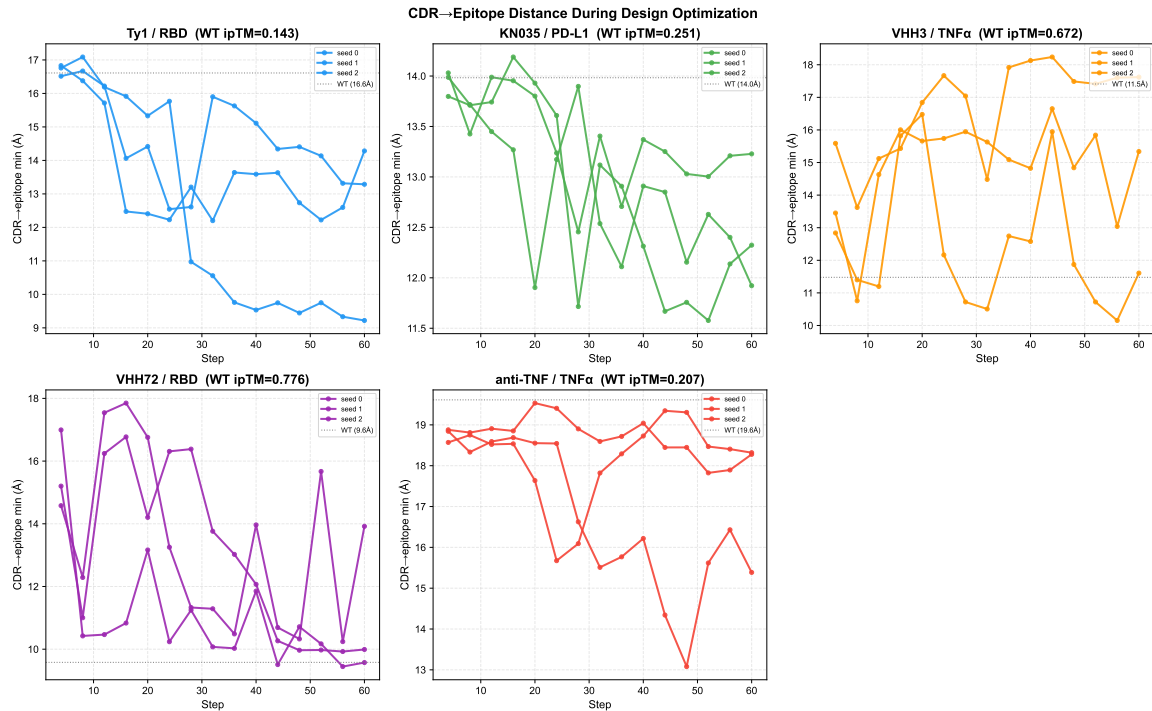


Figure 2. Design trajectories: CDR-to-epitope distance during optimization. CDR-to-epitope minimum expected distance (Å) vs. optimization step for all five targets, three independent seeds each. Best seed per target in bold color; other seeds in lighter opacity. Gray dotted line: WT cdr→epi. Lower is better. Ty1 and PD-L1 show consistent monotonic improvement. VHH72 and VHH3 show minimal change. anti-TNF shows early improvement that plateaus. The inter-seed spread for Ty1 (9.2–14.3 Å) underscores the necessity of multi-seed evaluation.

2.4 Multi-seed diversity necessitates replicate runs

Three independent random seeds per target (Fig. 2) revealed substantial inter-seed variation. For Ty1/RBD, seed 1 achieved cdr→epi = 9.2 while seeds 0 and 2 reached 14.3 and 13.3—a 5.1 spread. For PD-L1, all three seeds converged to similar ipTM (0.447–0.459) with distinct CDR sequences, suggesting a flat fitness landscape. For VHH3, seed 0 achieved cdr→epi = 11.6 while seed 2 reached only 15.3. These results demonstrate that single-seed evaluation can mislead—our earlier work on the B5 target showed Spearman $\rho = 0.486$ ($p = 0.329$, not significant) between single-seed and multi-seed median rankings. **We recommend a minimum of three seeds for any new target.**

2.5 RMSD cross-validation confirms pose preservation

A central concern is whether improved ipTM reflects genuine interface improvement or pose drift. We performed Kabsch cross-validation: folding WT and best-design sequences with full ESMFold2, Kabsch-aligning predicted target CA to crystal target CA, and computing binder RMSD under the same transform (Table 3). For the two improved targets, the change in binder RMSD was small ($\Delta = -1.45$ Å for Ty1, $+1.81$ Å for PD-L1), confirming pose preservation. For strong-binder controls with near-crystal poses (VHH72 bRMSD 3.22, VHH3 bRMSD 4.41), design maintained accuracy ($\Delta < 0.6$). For anti-TNF, both WT (44.39) and design (37.16) were far from crystal—the framework’s predicted pose is fundamentally incorrect, consistent with our

142 earlier finding that ESMFold2’s pose is a deterministic function of the framework sequence[32].

Table 3. Kabsch cross-validation against crystal structures. Predicted target CA aligned to crystal target CA; same transform applied to binder. Δ bRMSD: design – WT.

Target	WT ipTM	Design ipTM	WT bRMSD (Å)	Design bRMSD (Å)	Δ bRMSD (Å)
Ty1 / RBD	0.167	0.682	22.80	21.35	−1.45
KN035 / PD-L1	0.255	0.484	24.44	26.25	+1.81
VHH3 / TNF α	0.639	0.653	4.41	5.00	+0.59
VHH72 / RBD	0.786	0.745	3.22	3.31	+0.09
anti-TNF / TNF α	0.110	0.098	44.39	37.16	−7.23

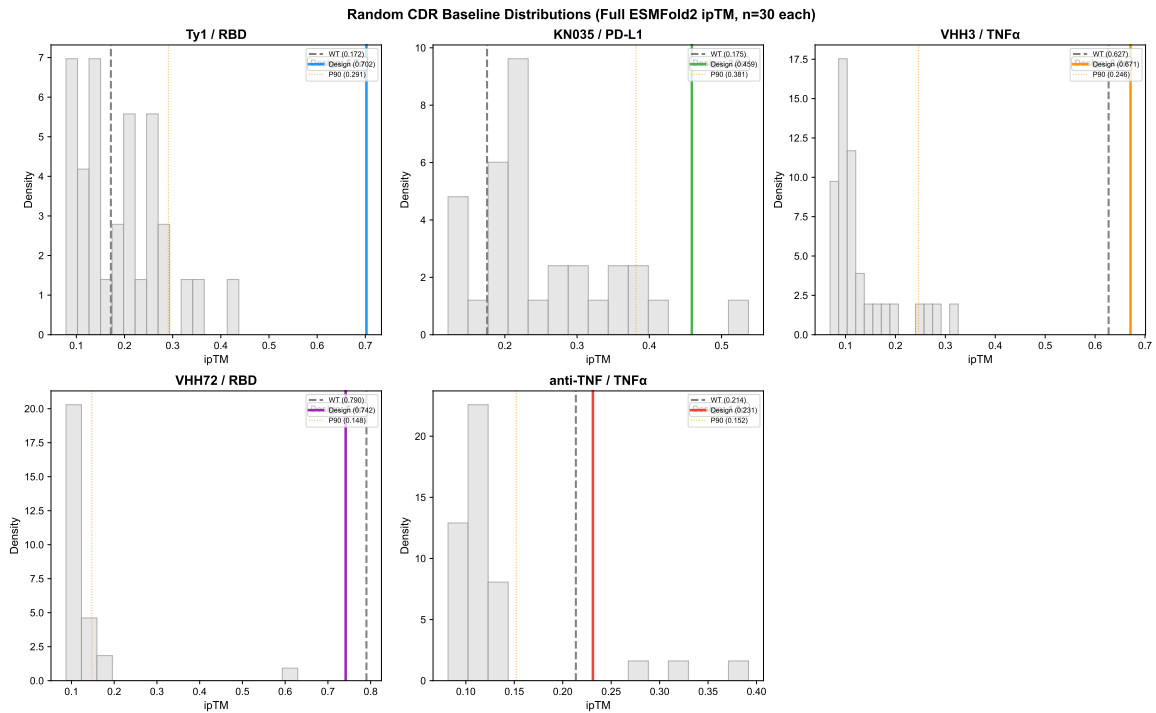


Figure 3. Random CDR baseline distributions establish statistical significance. Full ESM-Fold2 ipTM distributions for $n = 30$ random CDR sequences per target (gray histogram). Vertical dashed line: WT ipTM. Colored solid line: best design ipTM. Orange dotted line: 90th percentile of random distribution. $z\sigma$: design ipTM relative to random mean. Ty1 design (5.7σ) and PD-L1 design (2.2σ) significantly exceed random expectation. VHH3 and VHH72 WT ipTM already far above random, leaving no optimization headroom. anti-TNF design is within random distribution range.



Figure 4. Designed CDR sequences for all five targets. Sequence logo representation of the best designed CDR for each target. Positions differing from wild-type are highlighted in red. Right annotations show WT ipTM $\rightarrow$ Design ipTM, Δ , and number of mutations. Ty1 (11/22 mutated, 50%) and VHH3 (17/33 mutated, 52%) show the most extensive changes. PD-L1 (7/32, 22%) preserves structural disulfide cysteines. VHH72 (11/29, 38%) changes H1 and H2 while largely preserving H3. anti-TNF (10/19, 53%) undergoes extensive mutation but achieves minimal ipTM gain due to poor initial pose.

2.6 Determinants of design success

Analysis across the five targets reveals two principal determinants of design success:

CDR length. The two improved targets have ≥ 22 CDR positions (Ty1: 22, PD-L1: 32). anti-TNF has only 19 positions, providing insufficient degrees of freedom. Longer CDRs—particularly the elongated H3 loops characteristic of nanobodies—provide both a larger optimization surface and greater conformational flexibility.

Initial pose quality. VHH72 and VHH3, with WT ipTM > 0.65 and predicted poses within 5 of crystal, are already near-optimal. anti-TNF, with a 44 pose error, cannot be rescued. Ty1 and PD-L1 occupy the “sweet spot”: weak WT ipTM but correct pose basin (CDRs positioned near the epitope), enabling substantial improvement through CDR optimization alone. This suggests a practical two-stage workflow: (1) screen framework sequences to identify those in the correct pose basin, then (2) optimize CDRs of promising frameworks.

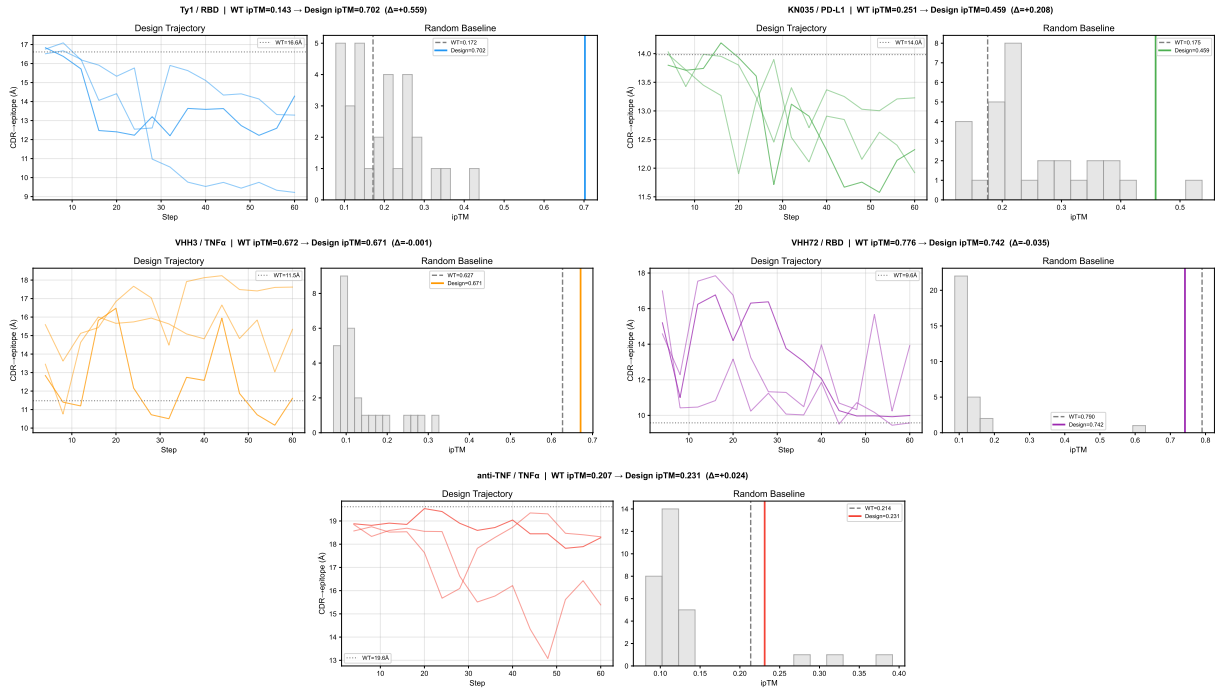


Figure 5. Extended Data Figure 1: Per-target design trajectories and random baseline distributions. For each target, left panel: CDR-to-epitope distance vs. optimization step (three independent seeds), right panel: random CDR baseline ipTM distribution with WT and design marked. (a) Ty1/RBD—best design achieves 9.2 cdr→epi and ipTM 0.702, 5.7σ above random. (b) KN035/PD-L1—design achieves ipTM 0.459, 2.2σ above random, limited by disulfide constraint. (c) VHH3/TNF α —strong WT binder, design preserved ipTM. (d) VHH72/RBD—strong WT binder, design preserved ipTM. (e) anti-TNF/TNF α —marginal improvement despite extensive CDR mutation; poor initial pose limits success.

3 Discussion

We have demonstrated that epitope-targeted nanobody CDR design can be performed rapidly—**under ten minutes per target on consumer hardware**—through gradient-based optimization of the ESMFold2 distogram. Our method achieves statistically significant ipTM improvements on weak binders (up to +0.559 for Ty1/RBD), preserves strong binders, requires no experimental screening or explicit docking, and provides multiple diverse candidate CDR sequences.

Our approach differs fundamentally from existing methods (Table 4): inverse folding[7, 8] requires a pre-specified backbone; RFdiffusion[10] generates backbones then sequences via ProteinMPNN; DiffAb[14] and dyMEAN[15] use diffusion or equivariant networks requiring GPU clusters. In contrast, our method directly optimizes CDR sequences over a continuous relaxation, using a single ESMFold2-Fast forward pass per gradient step—**60 forward/backward passes of a 721M-parameter model, tractable on a single consumer GPU or Apple MPS device**. This speed enables rapid iteration and high-throughput framework variant screening.

Table 4. Comparison with existing computational binder design methods.

Method	Epitope-specific	GPU time	Differentiable	No backbone required
ProteinMPNN[7]	No	<1 min	No	No
ESM-IF[8]	No	<1 min	No	No
RFdiffusion[10]	Partial	hours–days	No	Yes
DiffAb[14]	Yes	hours	Yes	Yes
dyMEAN[15]	Yes	hours	Yes	Yes
EasyNano (this work)	Yes	<10 min	Yes	Yes

The critical role of the wild-type logit bias merits emphasis. At first glance, one might expect the epitope loss to dominate optimization regardless of initialization. In practice, gradient signal from epitope loss is weak early in optimization (CDRs are far from the epitope and the distogram is uncertain), and the initialization bias must be tuned to allow exploration without enabling chaotic drift. We found $\beta = 2.0$ to be optimal, but this may require per-target adjustment for unusual CDR compositions or lengths.

Several limitations should be acknowledged. First, the method inherits ESMFold2’s pose accuracy: if the framework’s predicted pose is incorrect (as with anti-TNF, 44 from crystal), CDR optimization alone cannot compensate. Incorporating framework micro-tuning—single-residue mutations at framework positions flanking CDRs, which we have shown can shift the pose basin[32]—may address this. Second, the Fast model’s ipTM values are poorly calibrated (Fast-to-Full cdr→epitope discrepancies of up to 7), requiring full model evaluation for final ranking. Third, we have not yet experimentally validated designed sequences; binding must be confirmed by SPR or BLI. Fourth, the method currently optimizes only CDR sequences, not CDR lengths.

Future work should pursue: (1) experimental validation of top Ty1/RBD and PD-L1 designs; (2) incorporating ipTM as a direct loss term; (3) joint CDR length and sequence optimization; (4) framework residue co-optimization; (5) extension to multi-specificity design; and (6) application to other programmable scaffolds such as DARPin[33] and monobodies[34].

In conclusion, differentiable distogram optimization provides a fast, epitope-specific, and computationally efficient approach to nanobody CDR design. We anticipate applications in rapid prototyping of nanobody candidates against therapeutic targets, and expect the general principle—using structure prediction models as differentiable oracles for sequence optimization—to generalize to other protein design tasks.

4 Methods

4.1 Target preparation

Five target-framework pairs were selected from the PDB: Ty1 nanobody with SARS-CoV-2 spike (6ZXN, chains D/A), KN035 with PD-L1 (5JDS, B/A), VHH72 with SARS-CoV-2 RBD (6WAQ, A/B), anti-TNF VHH with TNF α (5M2J, D/A), and VHH3 with TNF α trimer (5M2M, D/B). For 6ZXN (spike trimer, 1060 residues per chain), the RBD domain (residues 330–535, 204 residues) was extracted. Epitope residues were target residues within 8 C α –C α of any nanobody

heavy atom, identified via Biotite[30]. CDR positions used Chothia numbering via abnumber[31].

4.2 Structure prior

Full ESMFold2 (1.3B)[18] with 3 recycling loops, 14 sampling steps, 4 diffusion samples. CA coordinates (averaged across diffusion samples) were discretized into 64 distance bins (2 to 22, tolerance 2.5).

4.3 Differentiable design loop

CDR residue logits $\mathbf{L} \in \mathbb{R}^{n_{\text{CDR}} \times 20}$ initialized Gaussian ($\sigma = 0.5$) + WT bias $\beta = 2.0$. Temperature: $T_t = 0.1 + 0.9 \cdot \frac{1}{2}[1 + \cos(\pi t/60)]$. ESMFold2-Fast (721M, 1 loop, 5 sampling steps). Adam optimizer (lr=0.05). Framework positions pinned after each step. Sixty optimization steps.

4.4 Evaluation

Full ESMFold2 (3 loops, 14 sampling steps, 1 diffusion sample, confidence head enabled). Three independent seeds per target. Random CDR baselines: $n = 30$ per target, amino acids sampled from natural frequencies (Cys excluded). Kabsch RMSD: predicted target CA aligned to crystal target CA; binder RMSD computed under same transform.

4.5 Compute

Apple Mac (Apple Silicon, MPS backend). ESMFold2 models in float32. MPS fallback enabled for unsupported operations. Design loop: ~ 6 s/step. Full pipeline per target (3 seeds + 30 random baselines): approximately 2.5 hours.

Data availability

All PDB structures from RCSB PDB (<https://www.rcsb.org/>): 6ZXX, 5JDS, 6WAQ, 5M2J, 5M2M. ESMFold2 and ESMC weights from EvolutionaryScale (<https://github.com/evolutionaryscale/esm>). Source data for all figures and tables are provided as Supplementary Data.

Code availability

EasyNano is available at [https://github.com/\[organization\]/EasyNano](https://github.com/[organization]/EasyNano) under MIT license. Frozen version archived at Zenodo (DOI to be provided).

Acknowledgements

We thank EvolutionaryScale for releasing ESMFold2 model weights. Supported by Qilu University of Technology (Shandong Academy of Sciences) and Shandong Provincial Hospital.

Author contributions

Y.H. conceived the study, developed the methodology, wrote all computational pipelines, performed experiments, and drafted the manuscript. W.C. contributed to data collection and computational experiments. J.W. contributed to target preparation, epitope analysis, and data interpretation. Y.L. provided clinical domain expertise, target selection guidance, and contributed to the manuscript. All authors reviewed and approved the final manuscript.

Competing interests

The authors declare no competing interests.

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